MUHAMMAD IMRAN | **BIO301 - Essentials of Genetics**

5th January 2021 | **MID TERM GRAND QUIZ**

Mid term Quiz will cover topics 1-103 (Introduction to genetics(topic 1)-epigenetics conclusion(topic103).

1. A male bee sex is _____ **Haploid**
2. In humans, _____ of somatic cells_____ **~ 220 types**
3. Shepherd's purse has two shape heart and narrow. If you cross a true breeding heart to a true breeding narrow, F1 will give _____ **Narrow**
4. The diameter of condensed scaffold associated chromatin is_____ **11 nm**
5. Which of the following is not the part of Interphase in the cell cycle?_____ **M phase**
6. Which of the following is true about phosphodiester linkage?_____ **5'-phosphate group of one nucleotide unit is joined to the 3'-hydroxyl group of the next nucleotide**
7. What is the composition of nucleotide?_____ **a base + a sugar + phosphate**
8. Which one of the following statement is correct about human genes?_____ **30% genes in humans are di-allelic and they exist in two forms**
9. The diameter of chromatin is_____ **11 nm**
10. The diameter of DNA nucleosome on the chromosome scaffold is_____ **2 nm**
11. Alleles separate during _____and reunite during _____._____ **Gametogenesis, Fertilization**
12. _____ states that the two alleles of a gene found on each of a pair of chromosomes separate independently of one another into sex cells._____ **Mendel's law of segregation**
13. How many most common chromosomal banding types are present?_____ **4**
14. The haploid cells of eukaryotes are usually known as _____ **Gametes**
15. Which one of the following is show polygenic pattern of inheritance?_____ **All of the given**
16. How many primary molecular mechanisms of epigenetics are identified?_____ **Two**
17. E.coli has about _____ **genes are organized into its one circular** chromosome_____ **3000 to 4000**
18. Pepper colour is controlled by _____ **Two genes**
19. ABO blood system is controlled by_ _____ **A tri-allelic gene**
20. Which one of the following statement is not correct about Co-dominance?_____ _____ **Both alleles are expressed in a homozygous individuals**
21. Which one is not regulate gene expression?_____ **Nucleotide modification**
22. Darker bands are generally found near the centromere or on the ends (telomeres) of the chromosome. The position of the dark-staining are _____or _____ **Heterochromatic region, heterochromatin**
23. In case of two gene interaction, the gene which is masking the expression of another is called _____ **Recessive**
24. Which of the following is not a common mode of histone modification in eukaryotes?___ _____ **Sulphonation**
25. Mitotic apparatus helps in....._____ **Chromosome movement**
26. The giant chromosomes found in _____and they may be as long as 300 µ and up to 10 µ in diameter._____ **Diptera**
27. The nucleosome is composed of a core of _____ **histone proteins and DNA** _____bp wrapped around them._____ **Eight, 147**
28. Name the phenomenon which can break the occurrence of linkage?_____ **Crossing over**
29. _____trait is a stable and heritable phenotype resulting from changes in chromosome without alterations in the DNA sequence_____ **Epigenetic**
30. Which one of the fur color of rabbit is controlled by maximum numbers of genotypes?_____ **Chinchilla**
31. _____ is an example of acquired variation_____ **Athletic skills**
32. Which one of the following statement is not correct about genome characteristics of prokaryotes?____ **Prokaryotes have less genome size as compared to eukaryotes**

33. Inherited variations result from the_____ **Activity of genes**

34. Which of the following is not the property of Garden Pea plant that was used by Gregor Mendel for his experiment?_____ **Long generation time**

35. Which comb shape is appeared when the chicken is homozygous recessive in both the R and P locus?_____ **Walnut Combs**

36. Which sub discipline of the genetics is considered the heredity of traits in groups of individuals, but the traits of concern are determined by many genes simultaneously?_____ **Quantitative genetics**

37. Chromatin fibers undergo folding, coiling and supercoiling during _____so that the chromosomes become progressively thicker and smaller._____**Prophase**

38. Which of the following organism has highest numbers of chromosome?_____ **Horsetail**

39. In physical map, units of distance are expressed in_____ **Base pairs or Mbs**

40. When one characteristic expresses itself over the other known as_____ **Dominant**

41. The trait that does not show through in the first generation is called_____**Recessive**

42. In_____, sister chromatids separate and move along the kinetochore microtubules toward opposite ends of the cell._____ **Anaphase**

43. Building blocks of nucleic acids are _____ **Nucleotides**

44. In which of the cats the pattern observed in coat colour is due to environmental effect?____ _____**Siamese**